Equation (1.27): PINN Evaluation Report

Run status. CPU smoke test: pipeline verification only; this is not a trained or validated solution.

1 Target and scope

The target is the full nonlinear equation

$$iE_t + (1 - i\epsilon_1\partial_z)(E_{xx} + E_{yy}) - E_{zz} - i\epsilon_2 E_{zzz} + (1 + i\epsilon_1\partial_z)(|E|^2 E) = 0.$$

The finite periodic box is a computational approximation, not an assertion of whole-space accuracy. All comparisons below use saved numerical outputs. A Fourier reference must itself pass grid, domain, and time-step checks before it can validate the PINN. The equation's linear weighted-decay estimate is not imposed as a nonlinear decay law.

2 Field comparison

The complex-field error includes phase disagreement. Peak and integrated squared-magnitude errors are distinct diagnostics; no phase alignment or renormalization is performed in these comparisons. Reference and PINN use the same evaluation coordinates.

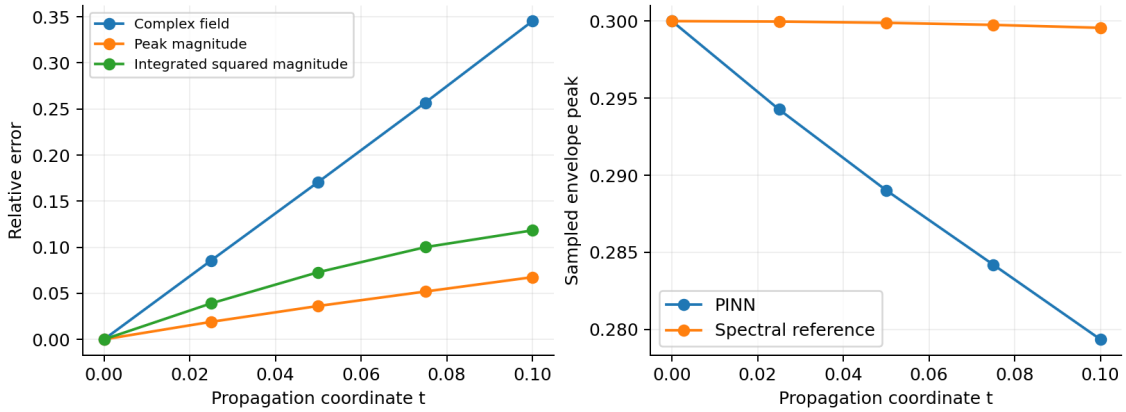


Figure 1: Measured PINN errors and sampled envelope peaks at the saved evaluation times. A falling predicted peak is not evidence of physical decay if the field errors are large.

t	Relative field error	Relative peak error	Relative mass error
0	2.548e-16	1.85e-16	0
0.025	0.08564	0.01903	0.03908
0.05	0.1708	0.03624	0.07287
0.075	0.257	0.05192	0.1001
0.1	0.3453	0.0675	0.1183

3 Spatial structure

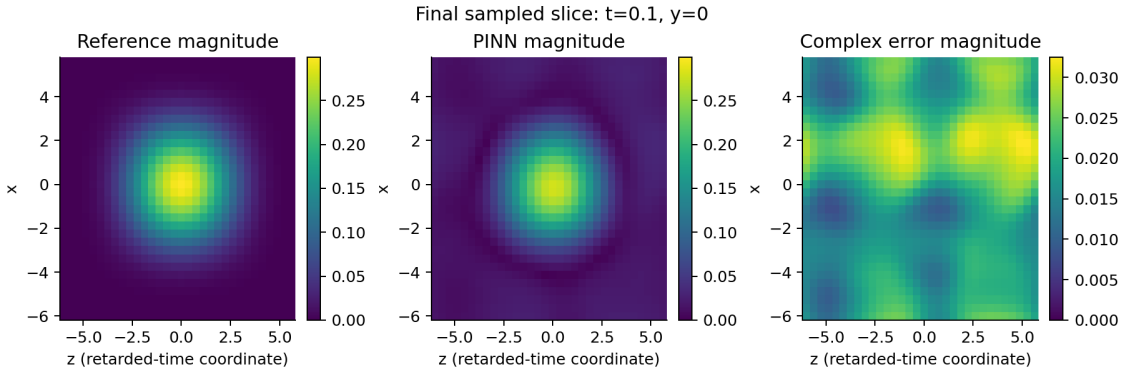


Figure 2: Final central transverse slice. Reference and prediction magnitudes share one color scale; the complex error has its own scale. The horizontal z axis is the paper's normalized retarded-time coordinate.

4 Interpretation and next checks

A successful smoke run establishes that derivatives, training, checkpointing, reference generation and reporting can execute. It does not establish a useful trained network. Evaluate multiple seeds and points withheld from training; inspect phase and peak errors; refine the reference and evaluate the PDE residual independently. Extend propagation only after short-window reference agreement. For inverse inference of the small coefficients, a separate identifiability and noise study is needed.

5 Recorded evaluation metadata

- Configuration: epsilon1=0.00831834509, epsilon2=0.00498719622; T=0.1; periods=[12.0, 12.0, 12.0]; initial amplitude=0.3, trigonometric power=2; dtype=float64.
- checkpoint_path: checkpoint.pt
- reference_path: reference.npz
- checkpoint_sha256: d8a79a57564d8920f9791b08734253f25498edc48df53fb79538f10bf9e1a808
- reference_sha256: acaa75206deda798c19a1a9b985b1d312eff665d784ef70eccabce00fec0f213
- max_pinn_relative_mass_drift: 0.11828231009745054
- max_reference_relative_mass_drift: 3.2124589401426532e-15
- equation: Dysthe3D-1.27
- checkpoint_step: 5
- max_relative_l2: 0.34532530961149954
- max_relative_peak_error: 0.06749635660165428
- max_mass_relative_error: 0.11828231009745338
- reference_convergence_status: not_established_by_this_comparison

The complete machine-readable metrics are supplied alongside this report. The source checkpoint and reference paths identify the inputs; preserve those artifacts and the training configuration for reproducibility.